QUANT

Mensuration & Geometry

(Circles, Quadrilaterals & Trigonometry)

eBOOK

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Geometry -II E-Book

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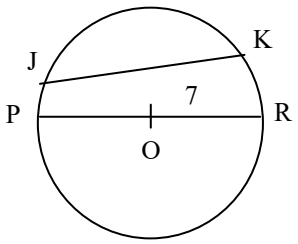
1.1 Circles

A circle is a set of points in a plane that are all located at the same distance from a fixed point (the center of the circle).

A **chord** of a circle is a line segment that has its endpoints on the circle. A chord that passes through the center of the circle is a diameter of the circle. A radius of a circle is a segment from the center of the circle to a point on the circle. The words "diameter" and "radius" are also used to refer to the lengths of these segments.

The **circumference** of a circle is the distance around the circle. If r is the radius of the circle, then the circumference is equal to $2\pi r$, where π is approximately 3.14. The area of a circle of radius r is is equal to πr^2.



In the circle above, O is the center of the circle and JK and PR are chords. PR is a diameter and OR is a radius. If OR = 7, then the circumference of the circle is 2π (7) = 14π, and the area of the circle is $\pi(7)^2$ = 49π.

Arc An arc is a part of a circle. A minor arc is an arc less than the semicircle and a major arc is an arc greater than a semicircle.

Central Angle An angle in the plane of the circle with its vertex at the centre is called a central angle.

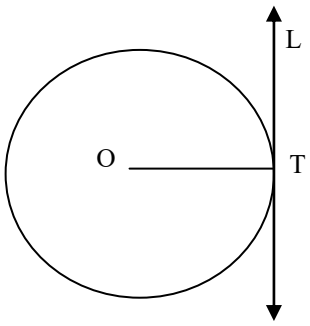
Measure of an arc

(i) The measure of a semicircle is 180°.
(ii) The measure of a minor arc is equal to the measure of its central angle.
(iii) The measure of a major arc = 360° – (measure of corresponding minor arc).

Congruent Circles: Circles with equal radii are called congruent circles.

Concentric Circles: Circles lying in the same plane with a common centre are called concentric circles.

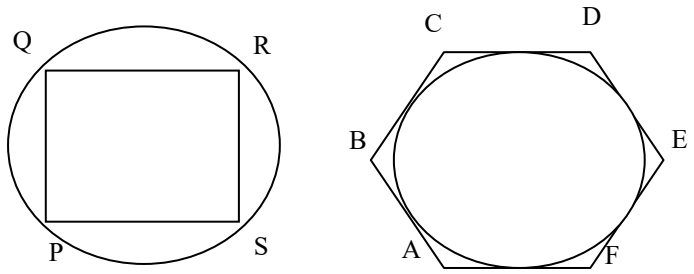
Tangent Circles: Circles lying in the same plane and having one and only one point in common are called tangent circles.



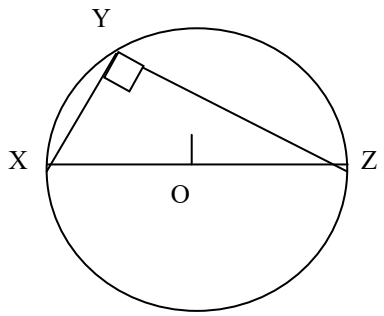
A line that has exactly one point in common with a circle is said to be tangent to the circle, and that common point is called the point of contact. A radius or diameter with an endpoint at the point of contact is perpendicular to the tangent line, and, conversely, a line that is perpendicular to a diameter at one of its endpoints is tangent to the circle at that endpoint.

The line L above is tangent to the circle and radius OT is perpendicular to L.

If each vertex of a polygon lies on a circle, then the polygon is inscribed in a circle and the circle is circumscribed about the polygon. If each side of a polygon is tangent to a circle, then the polygon is circumscribed about the circle and the circle is inscribed in the polygon.



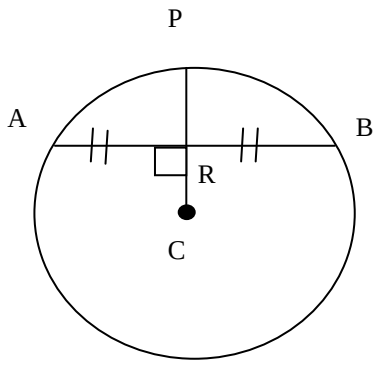
In the figure above, quadrilateral PQRS is inscribed in a circle and hexagon ABCDEF is circumscribed about a circle.



If a triangle is inscribed in a circle so that one of its sides is a diameter of the circle, then the triangle is a right triangle.
In the circle above, XZ is a diameter and the measure of angle XYZ is 90°.

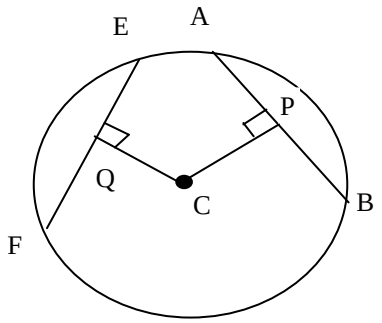
1.2 Some Important Properties of Circles

(i) The perpendicular from the centre of a circle to a chord of the circle bisects the chord.

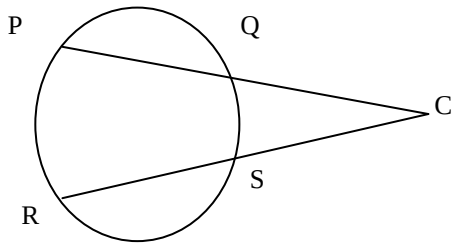


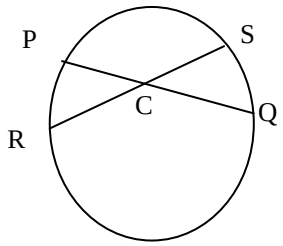
Conversely, the line joining the centre of the circle and the midpoint of a chord is perpendicular to the chord.

(ii) Equal chords of a circle or congruent circles are equidistant from the centre.

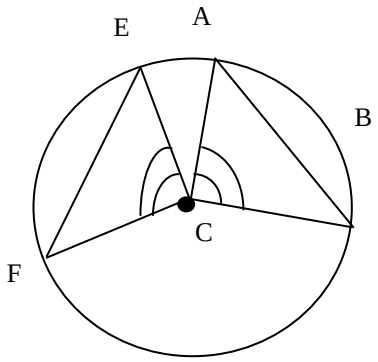


Conversely, two chords of a circle or congruent circles that are equidistant from the centre are equal.
Two chords PQ, RS intersect at a point then
CP × CQ = CR× CS



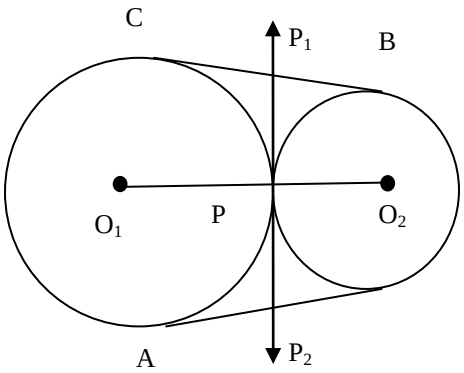


(iii) In a circle or congruent circles, equal chords subtend equal angles at the centre.

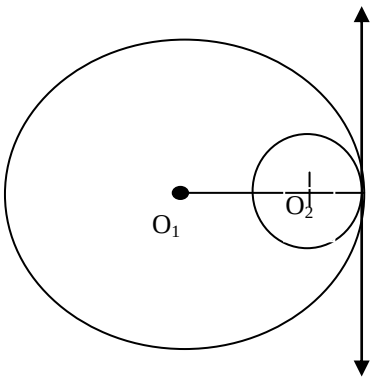


Conversely, chords, which subtend equal angles at the centre of the same or congruent circles, are equal.

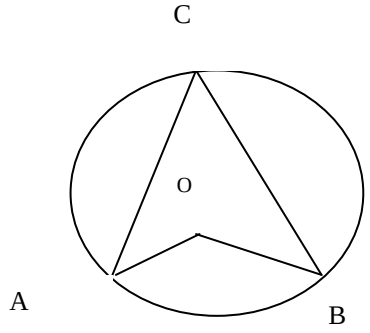
(iv) If the two circles touch each other externally, distance between their centres = sum of their radii.



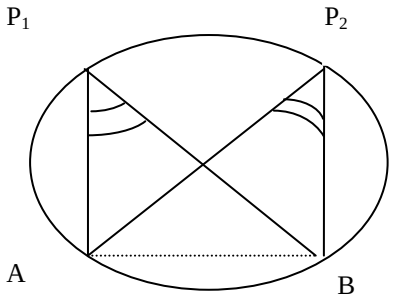
If the two circles touch each other internally, distance between their centres = difference of their radii.



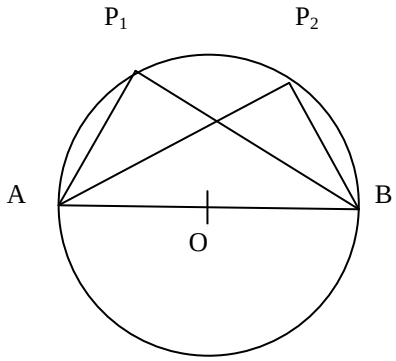
(v) The measure of an inscribed angle is half the measure of its intercepted arc. (Inscribed angle theorem) $\angle ACB = ½$ arc (ADB) = ½ $\angle AOB$.



Angles subtended is the same segment are equal

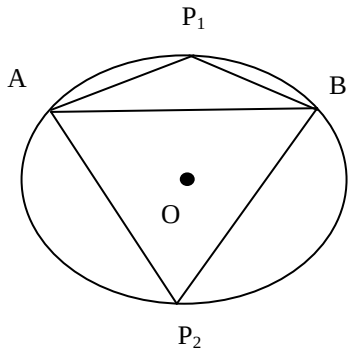


$\angle AP_1B = \angle AP_2B$.
Angle subtended in a semicircle is 90^0



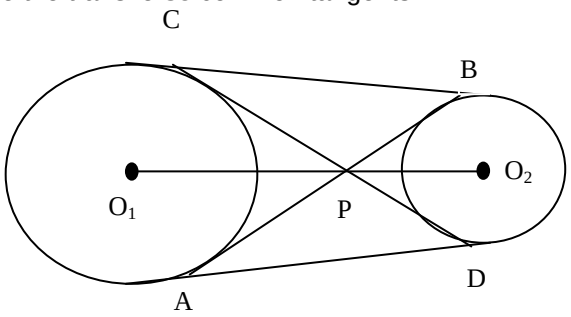
i.e $\angle AP_1B = 90^0 = \angle AP_2B$

Angle subtended in a minor segment is obtuse and that in a major segment is acute.



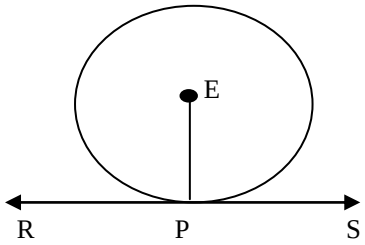
$\angle AP_1B > 90^0$ & $\angle AP_2B < 90^0$

(vi) For the two circles with centres A and B, PQ and RS are the direct common tangents, and CD and EF are the transverse common tangents.

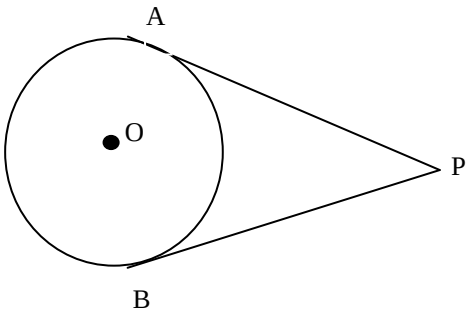


(Only two of both transverse common tangents and direct common tangents are possible.)

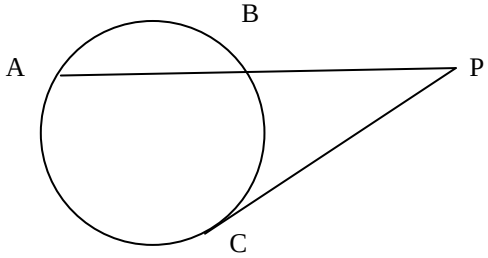
Tangent is always perpendicular to the line joining the centre and the point of tangency.



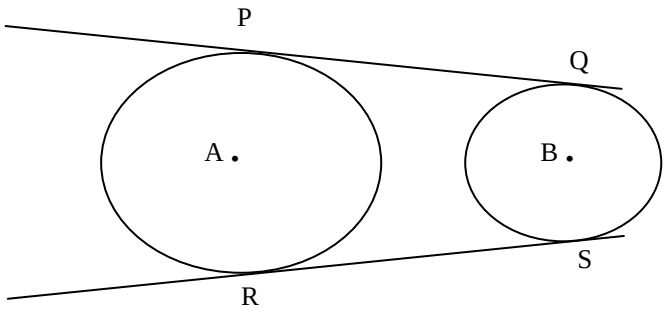
Two tangents from the same external point are equal in length. PA = PB



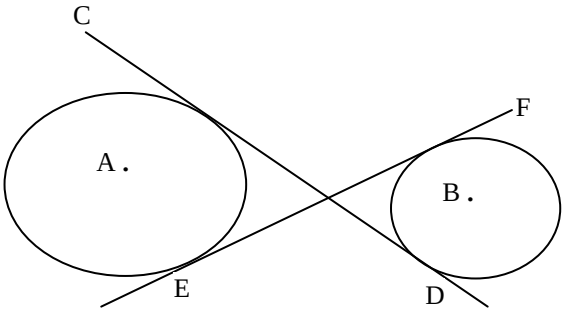
If AB is any chord of a circle and PC is the tangent (both for a external point P) than PA × PB = PC^2



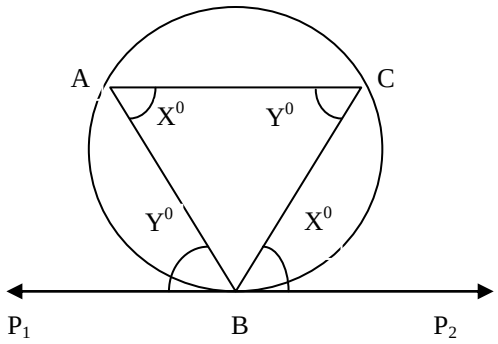
(vii) Length of direct common tangent = $\sqrt{(\text{distance between centres})^2 - (r_1 - r_2)^2}$



(viii) Length of transverse common tangent = $\sqrt{(\text{distance between centres})^2 - (r_1 + r_2)^2}$



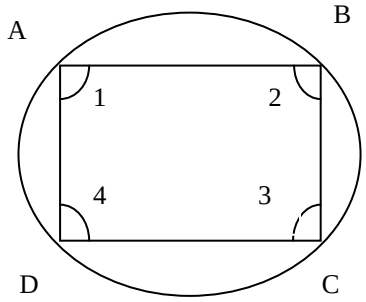
(ix) Alternate Segment Theorem:



Angle between any chord (at the point of tangency) and the tangent is equal to the angle subtended by the chord to any point on the other side of the segment (alternate segment).
In the fig $\angle CBP_2 = \angle CAB = X^0$ & $\angle P_1BA = \angle ACB = Y^0$

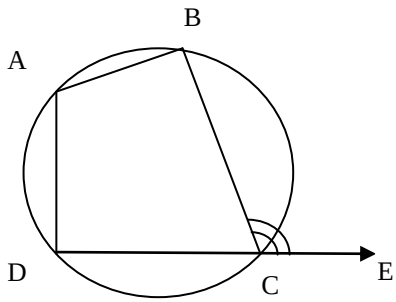
1.3 Cyclic Quadrilaterals

A quadrilateral is said to be cyclic if all its vertices lie on a circle. The points lying on a circle are called concyclic.

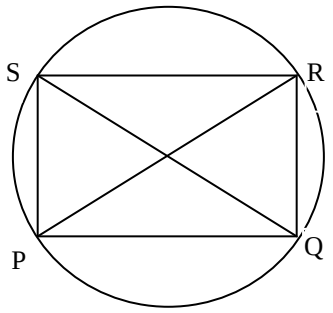


The opposite angles of a cyclic quadrilateral are supplementary.

Conversely, if the opposite angles of a quadrilateral are supplementary, then it is a cyclic quadrilateral.



An exterior angle of a cyclic quadrilateral is equal to the angle opposite to its adjacent interior angle. ∠BCE = ∠DAC.



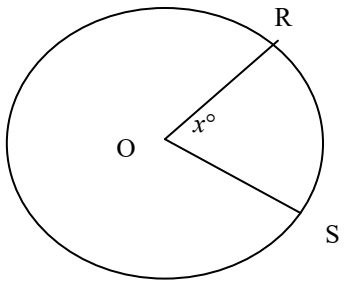
For any cyclic quadrilateral, the sum of product of two pairs of opposite sides equals the product of diagonals
PQ × RS + QR × SP = PR × SQ.

Area of a cyclic quadrilateral = $\sqrt{(s-a)(s-b)(s-c)(s-d)}$ where *s* is the semi perimeter and *a*, *b*, *c* and *d* are the sides of the quadrilateral.

Area of a cyclic quadrilateral in which a circle can be inscribed = $\sqrt{a \times b \times c \times d}$, where *a*, *b*, *c* and *d* are the sides of the quadrilateral.

1.4 Sectors of a Circle

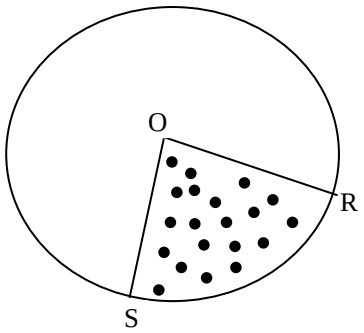
The number of degrees of arc in a circle (or the number of degrees in a complete revolution) is 360.



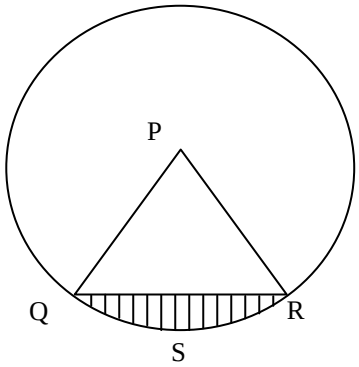
In the circle with center O above, the length of arc RS is *x*/360 of the circumference of the circle; for Ex, if *x* = 60°, then arc RS has length 1/6 of the circumference of the circle.

We can remember the following formulas:

- Length of arc RS = $2\pi r \times x/360$.
  ∵ The complete circle is having 360 degrees & any part of that shall be equal to *x*/360.



- Area of Sector ORS = $\pi r^2 \times x/360$.
  ∵ The complete circle is having 360 degrees & any part of that shall be equal to $x/360$.



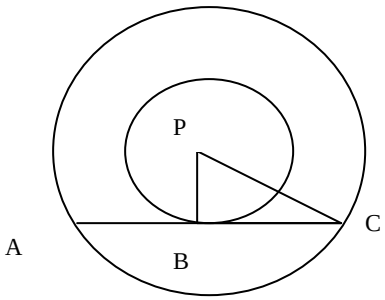
- Area of the segment of a circle (QSR) = $r^2\left[\frac{\pi\theta}{360°} - \frac{\sin\theta}{2}\right]$

  Perimeter of the segment = $2\pi\ r\left(\frac{\theta}{360°}\right) + 2r\sin\frac{\theta}{2}$

1.5 Example of Circles:

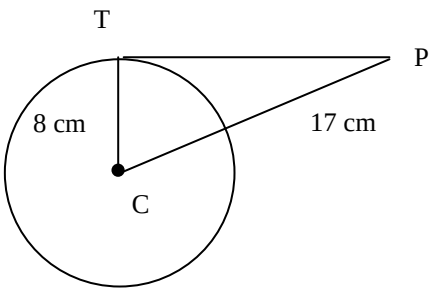
Ex. 1: Two concentric circles with centre P have radii 6.5 cm and 3.3 cm. Through a point A of the larger circle, a tangent is drawn to the smaller circle touching it at B. Find AC.

Sol: ∠PBC = 90° (A tangent is perpendicular to the radius at the point of contact)
So $(6.5)^2 = (3.3)^2 + (BC)^2$. So BC = 5.6. Hence AC = 2 × 5.6 = 11.2 cm.



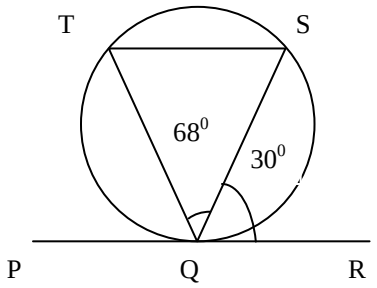
Ex. 2: Determine the length of the tangent to a circle of radius 8 cm from a point at a distance of 17 cm from the centre of the circle.

Sol: C is the centre of the circle P is a point outside the circle such that CP = 17 cm.



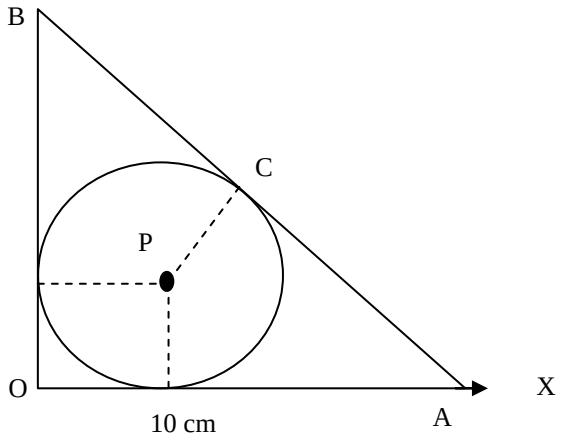
PT is a tangent from P to the circle ∴ CT = 8 cm.
Also CT is perpendicular to PT (radius is always perpendicular to the tangent at the point of contact)
∴ In rt. Angled Δ CTP, $PT^2 = CP^2 - CT^2 = 17^2 - 8^2 = 15^2$
∴ PT = 15 cm

Ex. 3: In the figure ∠ SQT = 68^0 and ∠ SQR = 30^0 find ∠ QST



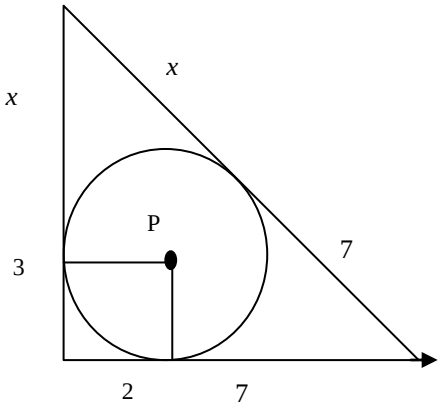
Sol: PQR is a tangent at Q. QS is a chord through Q.
Hence $\angle$ SQR = $\angle$ QTS = 30^0
(Angle between chord and tangent is equal to the angle is the alternate segment)
$\angle$ SQT + $\angle$ QTS + $\angle$ QST = 180^0
$\therefore$ $\angle$ QST = $180^0 - (30^0 + 68^0) = 82^0$

Ex 4. A circle of radius 3 cm is drawn touching both the axes OX and OY as shown is the figure. A tangent is drawn to the circle from a point A, 10cm away from O on OX, which meets OY at B. find the length of OB.



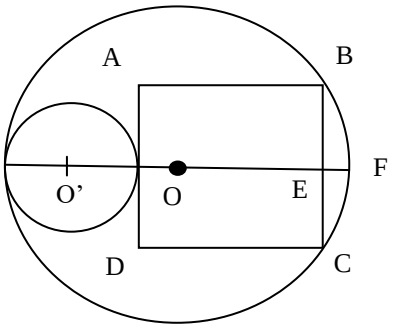
1. 10 cm
2. $\frac{20}{7}\sqrt{58}$ cm
3. $\frac{21}{2}$ cm
4. None of these

Sol:



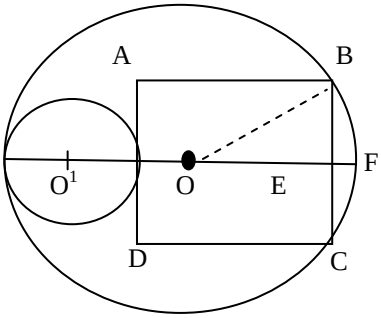
$(x + 7)^2 = 10^2 + (x + 3)^2$
(Tangents from the external point are equal)
$\Rightarrow x = 7.5$cms
$\Rightarrow$ OB = $x + 3 = 10.5$ cm

Ex 5. In the figure O and O' are the centers of the bigger and smaller circles respectively. The smaller circle touches the square ABCD at the midpoint of side AD. The radius of the bigger circle is 15 cm and the side of the square ABCD is 18 cm. Find the radius of the smaller circle.

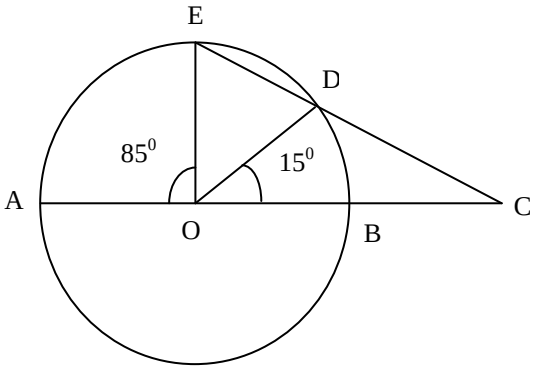


1. 4.25 cm 2. 4.5 cm 3. 4.45 cm 4. 5 cm

Sol. Radius of bigger circle - R
Side of square = $2a$
OE = R – EF = R – [$2R$– ($2r$ + $2a$)]
$OE^2 + EB^2 = OB^2$
i.e.$(2a + 2r - R)^2 + a^2 = R^2$, for a = 9 cm
& R = 15 cm
$\Rightarrow r$ = 4.5 cm



Ex 6. In the given figure, AB is the diameter of the circle with center O. If ∠ BOD = 15° & ∠ EOA = 85°, then find the value of ∠ ECA



1. 20^0 2. 25^0 3. 35^0 4. Can't be determined

Sol: ∠ EOA = 85^0, ∠ BOD = 15^0
∴∠ EOD = $180^0 - (85^0 + 15^0) = 80^0$
∴In Δ OED, OE = OD (radii)
∴∠ OED = ∠ ODE = 50^0
∴In Δ OEC, ∠ EOC = 80^0+15^0
= 95^0+∠ OEC=50^0
$\Rightarrow$ ∠ ECA = $180^0 - (95 + 50^0) = 35^0$

1.6 Quadrilaterals

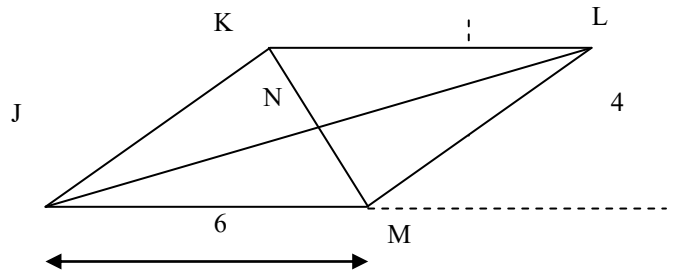
A polygon with four sides is a quadrilateral. In a quadrilateral, sum of all angles is 360°. Area of a quadrilateral = ½ × one of the diagonals × sum of the perpendiculars drawn to that diagonal from the opposite vertices.

The different kinds of quadrilaterals are parallelogram, rectangle, square, rhombus, trapezium and kite.

Parallelogram:

A quadrilateral in which both pairs of opposite sides are parallel is a parallelogram. The opposite sides of a parallelogram also have equal length. In a parallelogram opposite sides are parallel and equal. Opposite angles are equal. Diagonals bisect each other. Sum of any 2 adjacent angles = 180°.

A parallelogram inscribed in a circle is always a rectangle. Parallelogram circumscribed about a circle is always a Rhombus.

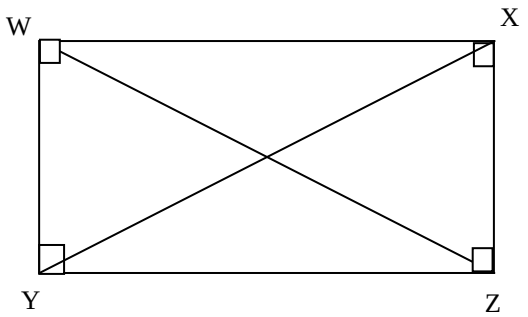


In parallelogram JKLM, JK || LM and JK = LM, KL || JM and KL = JM, The diagonals of a parallelogram bisect each other that is, KN = NM and JN = NL.
The area of a parallelogram is equal to: Base × height
The area of JKLM is equal to 4 x 6 = 24. Every diagonal of a parallelogram divides it into two triangles of equal area.
Parallelograms that lie on the same base and between the same parallels are equal in area.

Rectangle:

A parallelogram with right angles is a rectangle. In a rectangle, each pair of opposite sides is parallel and equal. Diagonals are equal and bisect each other, but not at right angles. A parallelogram is a rectangle if its diagonals are equal.

Perimeter of rectangle = 2(L + B), where L = Length, B = Breadth
Area of rectangle = LB.
Area is written in the square units of sides.
Diagonal^2 =L^2 + B^2
The perimeter of WXYZ = 2[3 + 4] = 14 and the area of WXYZ is equal to 3 × 4 = 12.
The diagonals of a rectangle are equal; therefore WY = XZ = √(9 + 16) = √25 = 5



Square:

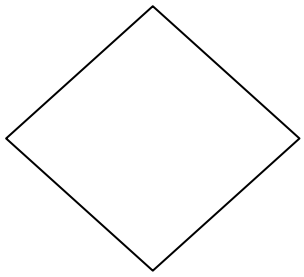
A rectangle with all sides equal is known as square. In a square, all 4 sides are equal. All the 4 angles are equal & each angle is equal to 90°. Diagonals are equal and bisect each other at right angles. The perimeter of a square is '4a' and the area of the square is 'a^2', where 'a' is the side of the square. Every square is a rhombus, rectangle and parallelogram.

When a square is inscribed in a circle, the diagonal is equal to the diameter of the circle. When a circle is inscribed in a square, side of the square is equal to the diameter of the circle.

Rhombus:

In a rhombus all the sides are equal and all the angles are not equal. In a rhombus, the two pairs of opposite sides are parallel. Diagonals are not equal but they bisect each other at right angles.

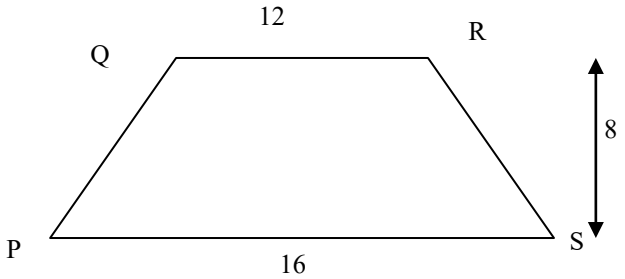
Opposite angles are equal.



Area = ½ d_1d_2, where d_1 & d_2 are two diagonals of a rhombus. $(\text{side})^2$ = $(\frac{1}{2}\text{ one diagonal})^2$ + $(\frac{1}{2}\text{ other diagonal})^2$ Every rectangle, square and rhombus is a parallelogram.

Trapezium:

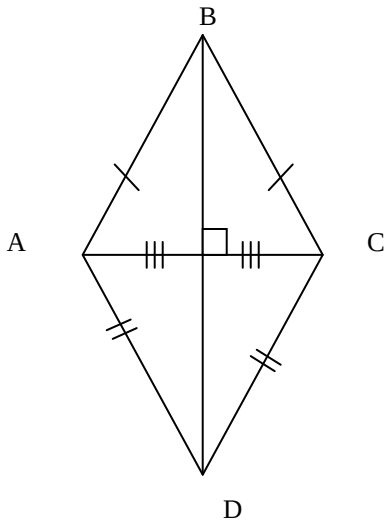
A quadrilateral with two sides that are parallel but the other two sides are not parallel, as shown below is a trapezoid.



The area of trapezoid PQRS may be calculated as follows:
½ × sum of parallel sides × height = ½ × (QR + PS)(8) = ½ × (28 × 8) = 112.

A trapezium inscribed in a circle is an isosceles trapezium. In an isosceles trapezium, the oblique sides (the sides which are not parallel) are equal. Angles made by each parallel side with the oblique side are equal.

Kite:



A quadrilateral is called a kite, if it has two pairs of equal and adjacent sides. In a kite, two pairs of adjacent sides are congruent. The diagonals of a kite intersect at right angles. The longer diagonal bisects the shorter diagonal.
Area = ½ × product of diagonals.

Solved Examples of Quadrilaterals

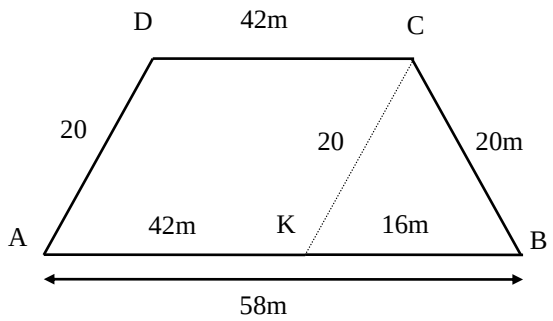
Ex 7. The parallelogram ABCD is composed of four congruent triangles, such that each triangle has two sides common with two of the other triangles. E is the point of intersection of diagonals. If BE is 3 & CE is 4, what is the perimeter of the entire parallelogram?

1. 15 2. 40 3. 20 4. 30 5. 50

Sol. As all the four triangles are congruent and the middle point where they all meet is consisting of 4 angles and each angle being equal, each angle becomes 90°.

The value of BE and CE is given to be 3 and 4 cm, thus the side of this rhombus becomes 5. The perimeter of it becomes 5 × 4 = 20,

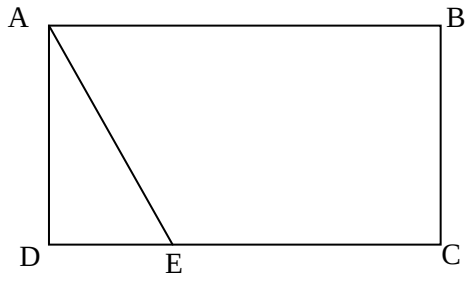
Ex 8. Find the area of a trapezium whose parallel sides are of lengths 58 metres and 42 m and whose non-parallel sides are equal; each being 20 m.



1. 840 sq. m 2. 1000 sq. m. 3. 916.50 sq. m 4. 800 sq. m

Sol. In triangle *CKB* height, $h = \sqrt{(20^2 - 8^2)} = 4\sqrt{21}$.
Area of trapezium = ½ × (sum of || sides)
(perp. Distance) = ½ × (42 + 58) ($4\sqrt{21}$) = $200\sqrt{21}$ = 916.50 sq. cm.

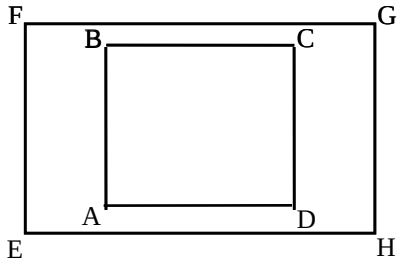
Ex 9. Referring to the diagram below, there is a rectangle ABCD. The area of isoceles triangle ADE is 7 sq cm. Also EC = 3(DE). What is the area of rectangle ABCD?



1. 21 2. 28 3. 42 4. 56

Sol. 1/2 × DE × AD = 7.
So DE × AD = 14.
Area (ABCD) = CD × AD.
Now CD = 4 DE.
Thus area will be 4 DE × AD ⇒ 4 × 14 = 56.

Ex 10. ABCD is a square, EFGH is a rectangle. AB = 3, EF = 4, FG = 6. The area of the region outside of ABCD and inside EFGH is



1. 15 2. 12 3. 9 4. 6

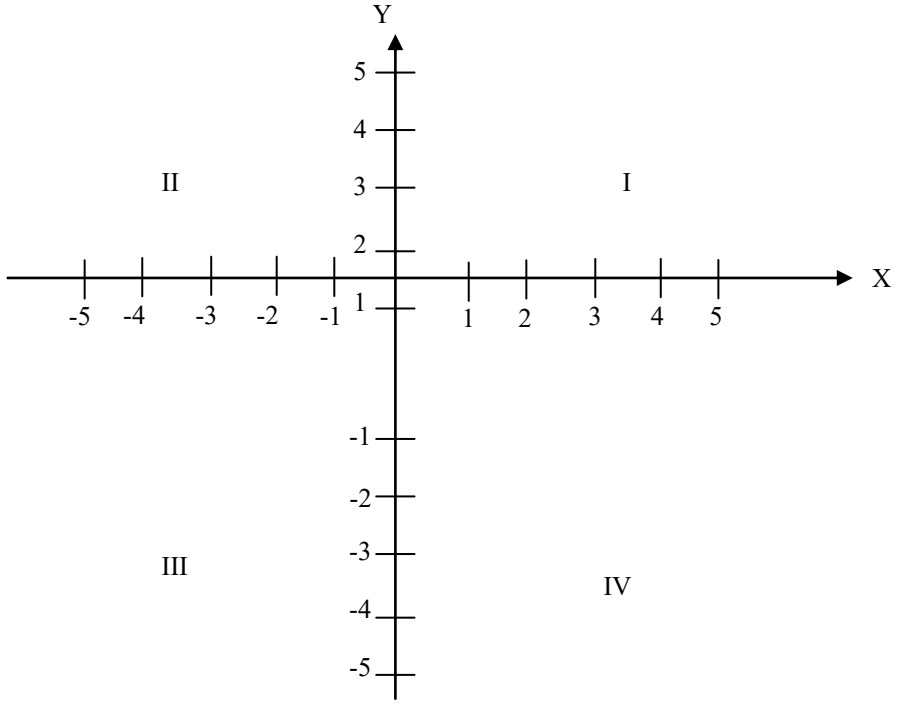
Sol. Area outside sq ABCD and inside square EFGH = 24 – 9 = 15 sq units.

Ex 11. An equilateral triangle BPC is drawn inside a square ABCD. What is the value of the angle APD?

1. 75° 2. 90° 3. 120° 4. 150°

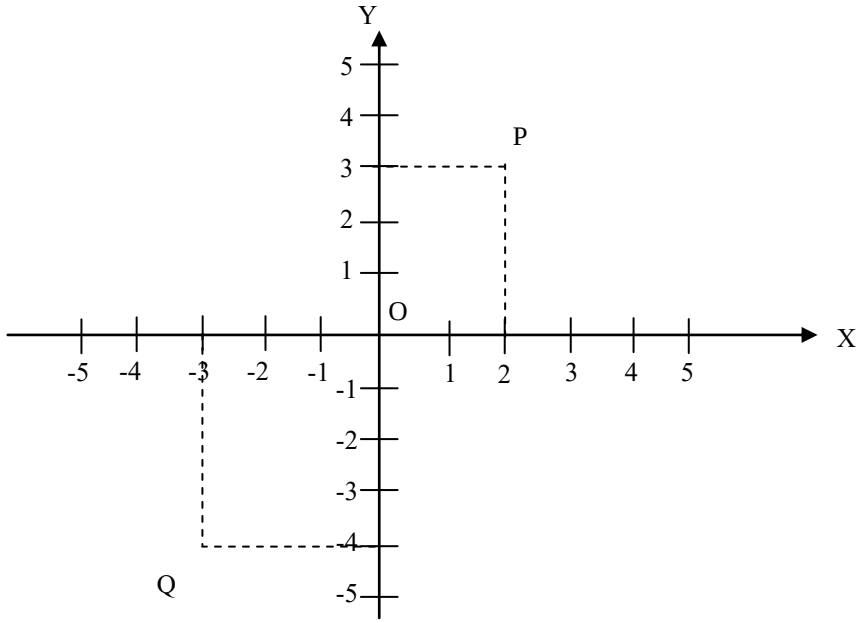
Sol: ΔBPC is equilateral.
⇒ BP = AB,
⇒ This means Triangle APB is isosceles
$\angle ABP = 30^\circ \Rightarrow \angle BAP = \angle APB = 75^\circ$
$\Rightarrow \angle DAP = \angle ADP = 15^\circ$
$\Rightarrow \angle APD = 180 - (15+15) = 150^\circ$

2.1 Coordinate Geometry



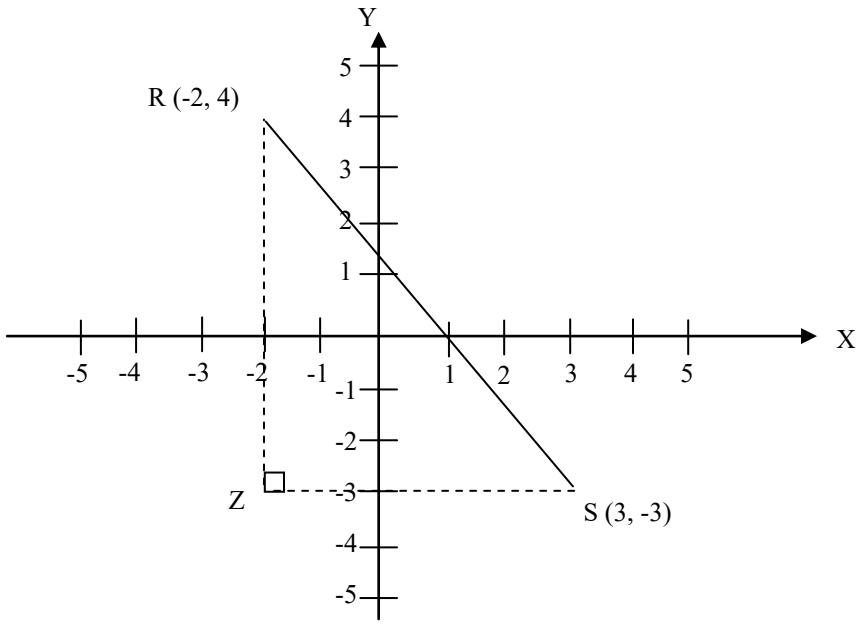
The figure above shows the (rectangular) coordinate plane. The horizontal line is called the *x*-axis and the perpendicular vertical line is called the *y*-axis. The points at which these two axes intersect, designated O, is called the origin. The axes divide the plane into four quadrants, I, II, III and IV, as shown.

Each point in the plane has an *x*-coordinate and a *y*-coordinate. A point is identified by an ordered pair (*x*, *y*) of numbers in which the *x*-coordinate is the first number and the *y*-coordinate is the second number



In the graph above, the (*x*, *y*) coordinates of point P are (2, 3) since P is 2 units to the right of the *y*-axis (that is, $x = 2$) and 3 units above the x-axis (that is, $y = 3$). Similarly, the (*x*, *y*) coordinates of point Q are (-3, -4). The origin O has coordinates (0, 0).

One way to find the distance between two points in the coordinate plane is to use the Pythagorean theorem.



To find the distance between points R and S using the Pythagorean theorem, first draw the triangle as shown. Note that Z has (x, y) coordinates (-2, -3), RZ = 7, and ZS = 5. Therefore, the distance between R and S is equal to $\sqrt{(7^2 + 5^2)} = \sqrt{74}$.

2.2 Important Formulas of Coordinate Geometry

Distance between two points:

Distance between two points P (x_1, y_1) and Q (x_2, y_2) is given by the following formula. PQ = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Ex 12. What is the distance between the two points P (2, - 1) and Q (-3, 1)?

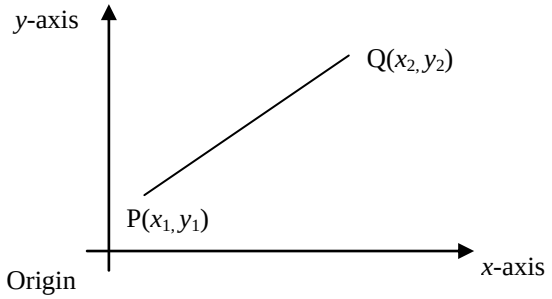
Sol : Apply the above formula, and get the answer to be $\sqrt{29}$.

Ex 13. Find the co-ordinates of a point on x-axis, which is at a distance of 5 units from the point (6, -3).

Sol : Let the co-ordinates of the point on the x-axis be (x, 0).
As distance = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$,
so 5 = $\sqrt{(x-6)^2 + (0-(-3))^2}$
$\Rightarrow 25 = x^2 - 12x + 36 + 9$.
$\Rightarrow x = 2$ or $x = 10$.
So the required points are (2, 0) and (10, 0)

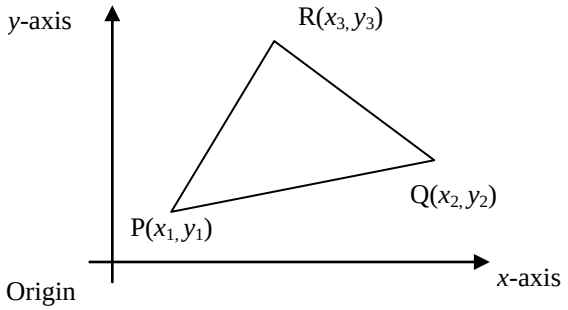
Slope of a line:

Slope of a line is the tangent of the angle which it makes with positive direction of the x-axis



The slope of the line joining the points P(x_1, y_1) and Q(x_2, y_2) is given by slope= $\frac{y_2 - y_1}{x_2 - x_1}$

Area of a Triangle:

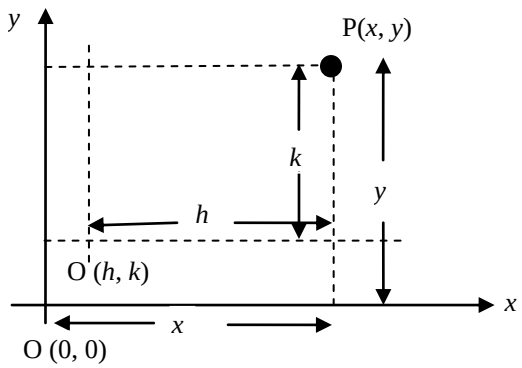


The area of a triangle with its vertices P(x_1, y_1), Q (x_2, y_2) and R(x_3, y_3) is given by

$$\Delta = \frac{1}{2}\left[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right]$$

If the area of the triangle is zero, it implies that the points are collinear.

Shifting of origin:



If the co-ordinates of a point P are (x, y) with reference to the origin O (0, 0). Now if the origin is shifted the co-ordinates of the point P with reference to the new origin will be ($x - h$, $y - k$)

Section Formula:

If P is a point dividing the join of two points A(x_1, y_1) and

B (x_2, y_2) internally in the ratio m: n, then the coordinates (x, y) of P are given by $\left[\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right]$.

2.3 More Solved Examples of Coordinate Geometry:

Ex 14. Find the co-ordinates of point Z, which divides the join of P (4, -5) and Q (6, 3) in the ratio 2 : 5.

Sol: Let the co-ordinates of point Z are (x, y).

Now x = $\frac{mx_2 + nx_1}{m+n} \Rightarrow \frac{(2\times 6)+(5\times 4)}{2+5} = \frac{32}{7}$.

And y = $\frac{my_2 + ny_1}{m+n} \Rightarrow \frac{(2\times 3)+\{5\times(-5)\}}{2+5} = \frac{-19}{7}$.

z = $\left(\frac{32}{7}, -\frac{19}{7}\right)$

If P(x, y) divides the join of A(x_1, y_1) and B(x_2, y_2) externally in the ratio $m : n$, then the coordinates of P are given by $\left[\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n}\right]$

If P is the mid point of the line joining the points A(x_1, y_1) and B(x_2, y_2), then coordinates of P are given by $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Ex 15. Find the co-ordinates of the mid-point of the line segment joining the points M (4, -6) and N (-2, 4).

Sol: Mid-point = $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) \Rightarrow \left(\frac{4+(-2)}{2}, \frac{-6+4}{2}\right)$ **=** (1, - 1).

Ex 16. Find the co-ordinates of the point of intersection of the medians of triangle MNO; given M = (-2, 3), N = (6, 7), O = (4, 1).

Sol: Point of intersection of medians is called as centroid, at which each median is divided in the ratio 2 : 1.

Firstly find the mid-point of NO and take it as P.

So P = $\left(\frac{6+4}{2}, \frac{7+1}{2}\right)$ = (5, 4).

Take the point C, where medians meet and it will divide the median MP in the ratio 2 : 1.

Co-ordinates of C = $\left[\left(\frac{(2\times 5)+(1\times -2)}{2+1}\right), \left(\frac{(2\times 4)+(1\times 3)}{2+1}\right)\right] = \left(\frac{8}{3}, \frac{11}{3}\right)$.

- The coordinates of the centroid of a triangle with vertices A(x_1, y_1), B (x_2, y_2) and C (x_3, y_3) are $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$
- The coordinates of the in-centre I (x, y) of the triangle with vertices A(x_1, y_1), B (x_2, y_2) and C (x_3, y_3) are $\left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c}\right)$ where a, b and c are the lengths of three sides opposite to $\angle$ BAC, $\angle$ ABC and $\angle$ ACB.

Conditions for a parallelogram: If A(x_1, y_1), B (x_2, y_2), C (x_3, y_3) and D (x_4, y_4) are four non collinear points such that $x_1 + x_3 = x_2 + x_4$ and $y_1 + y_3 = y_2 + y_4$, then quadrilateral ABCD is a parallelogram.

- Conversely, if A(x_1, y_1), B (x_2, y_2), C (x_3, y_3) and D (x_4, y_4) form a parallelogram, then $x_1 + x_3 = x_2 + x_4$ and $y_1 + y_3 = y_2 + y_4$.

Equation of a line: A geometrical condition satisfied by all points P (x, y) on a straight line when expressed in x and /or y is called the equation of a straight line.

Equation of lines parallel to the axes:

(i) The equation of a line parallel to the x axis and at a distance of k units above it is y = k. y = k is also called the y intercept.
(ii) The equation of a line parallel to the x axis and at a distance of k units below it is y = -k. y = -k is also called the y intercept.
(iii) The equation of a line parallel to the y axis and at a distance of k units to the right of it is x = k. x = k is also called the x intercept.
(iv) The equation of a line parallel to the y axis and at a distance of k units to the left of it is x = -k. x = -k is also called the x intercept.

Standard forms for the equation of a line:

(i) **The Point Slope form:** The equation of a straight line passing through the point (x_1, y_1) and having slope m is $y - y_1 = m(x - x_1)$.
(ii) **The Two Point form:** The equation of a straight line passing through the points (x_1, y_1) and (x_2, y_2) where $x_1 \neq x_2$ is $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$
(iii) **Slope Intercept form:** The equation of a line having slope m and making an intercept c on y axis is $y = mx + c$.
(iv) **Double Intercept form:** The equation of a line making intercepts a and b, when $a \neq 0$ and $b \neq 0$ on the x axis and y axis respectively is $\frac{x}{a} + \frac{y}{b} = 1$
(v) **The General form:** The most general form of an equation is of the form $Ax + By + C = 0$, where A, B and C are real numbers.

Parallel and Perpendicular Lines:

(i) Two lines whose slopes are m_1 and m_2 are parallel to each other, if and only if $m_1 = m_2$.
(ii) Two lines whose slopes are m_1 and m_2 are perpendicular to each other if and only if $m_1 \times m_2 = -1$. Thus, if the slope of a line is m, the slope of a line perpendicular to it is $-1/m$.

Point of intersection of two lines: If $A_1x + B_1y + C_1 = 0$ and $A_2x + B_2y + C_2 = 0$ are two straight intersecting lines, then coordinates of the point of intersection can be obtained by solving the equations simultaneously for x and y.

Angle between two lines: If the equations of the two lines are $A_1x + B_1y + C_1 = 0$ and $A_2x + B_2y + C_2 = 0$, then $\tan\theta = \frac{A_1B_2 - A_2B_1}{A_1A_2 + B_1B_2}$.

Perpendicular distance of a point from a line:

If $P(x_1, y_1)$ is any point and $Ax + By + C = 0$ is a line, then the perpendicular distance of P from the line is given by $\left|\frac{Ax_1 + By_1 + C}{\sqrt{A^2 + B^2}}\right|$.

If this point P is the origin, then perpendicular distance is given by $\left|\frac{C}{\sqrt{A^2 + B^2}}\right|$.

Perpendicular distance between two lines:
If $Ax + By + C_1 = 0$ and $Ax + By + C_2 = 0$ are two parallel lines, and the perpendicular distance between them is given by $\left|\frac{C_1 - C_2}{\sqrt{A^2 + B^2}}\right|$.

Condition of concurrency of three straight lines:
If $A_1x + B_1y + C_1 = 0$, $A_2x + B_2y + C_2 = 0$ and $A_3x + B_3y + C_3 = 0$ are three straight lines, then the three lines will be concurrent (i.e intersect at a point) if $A_3(B_1C_2 - B_2C_1) + B_3(C_1A_2 - C_2A_1) + C_3(A_1B_2 - A_2B_1) = 0$.

Ex 17. What is the slope of the line passing through the points J (-2, 3) and (2, 7)?

Sol: $\left[\frac{y_2 - y_1}{x_2 - x_1}\right] = \left[\frac{7-3}{2-(-2)}\right] = \frac{4}{4} = 1.$

Ex 18. Find the equation of the line whose slope is 2 and *y* intercept is – 3.

Sol: Given m = 2 and c = - 3. Substituting values in $y = mx + c$, we get $y = 2x - 3$.

Ex 19. Find the equation of the line passing through (2, -1) and parallel to the line $2x - y = 4$.

Sol: The given line is $2x - y = 4 \Rightarrow y = 2x - 4$ (Converting into the form of $y = mx + c$)
Its slope = 2. The slope of the parallel line should also be 2.
Hence for the required line
m = 2 and $(x_1, y_1) = (2, -1)$.
Equation = $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1} \Rightarrow \frac{y - y_1}{x - x_1} = m.$
$\Rightarrow y - y_1 = m(x - x_1) \Rightarrow y - (-1) = 2(x - 2)$
$\Rightarrow y = 2x - 5.$

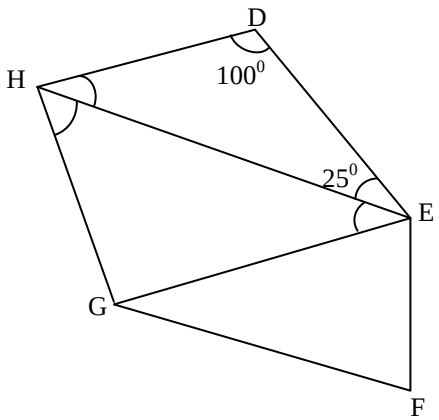
Ex 20. Find the equation of the line parallel to the line passing through (5,7) and (2,3) and having *x* intercept as -4.

Sol: Slope of the given line = (7 – 3) / (5 – 2) = 4/3. So the slope of the required line is also 4/3. One point on this line is (-4, 0). Hence the equation of the line is $y - 0 = 4/3\,(x + 4) \Rightarrow 3y = 4x + 16$.

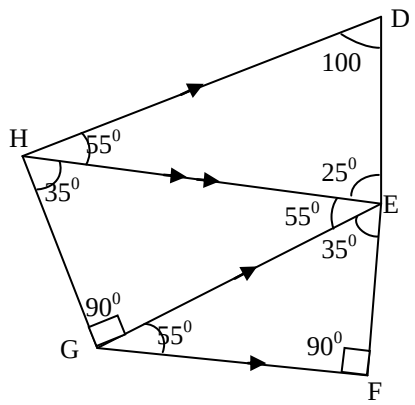
Ex 21. Find the coordinates of the circum-centre of the triangle whose vertices are (0,0) , (8,0) and (0,6). Find the Circum-radius also.

Sol: Circum-centre is the point of intersection of the perpendicular bisectors of the three sides of a triangle.
Let S (x, y) be the circumcentre. SA = SB = SC. $\therefore \sqrt{x^2 + y^2} = \sqrt{(x-8)^2 + (y-0)^2}$.
Also $\sqrt{x^2 + y^2} = \sqrt{(x-0)^2 + (y-6)^2}$.
Squaring $x^2 + y^2 = x^2 - 16x + 64 + y^2$. So $x = 4$.
Also $x^2 + y^2 = x^2 + y^2 - 12y + 36$.
So $y = 3$. Hence coordinates of the circumcentre is (4, 3). Circumradius = SA $= \sqrt{x^2 + y^2} = \sqrt{16+9} = 5$.

Ex 22. Find ∠ FGE, given that HD||GE and GF||HE. ∠DHG = 90^0 ∠DEG = 90^0



Sol:



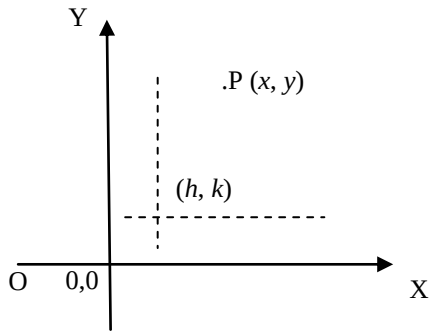
$\therefore \angle$ FGE = 55^0

Ex 23. A point P in the $x - y$ plane has the co-ordinates as (- 3, 6). If the origin is shifted to the point (5, 8) then find the co-ordinates of the point P with reference to the new origin.

Sol: For a point P(x, y) in the $x - y$ plane if the origin is shifted to new location at (h, k) than co-ordinates of the point with reference to the new origin are

$\equiv$ (- 3 – 5, 6 – 8)

$\equiv$ (- 8, - 2)



Ex 24. Find the equation of the straight line through the points (2, 3) and (5, 6).

Sol: The equation of the straight line through point P(x_1 ,y_1) & Q(x_2 ,y_2) is $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

i.e. (y – 3) = $\frac{6-3}{5-2}(x-2)$ i.e. y – 3 = x – 2 or y = x + 1

Ex 25. Find the equation of straight line passing through (2, 3) and perpendicular to the line 3x + 2y + 4 = 0

Sol: The given line is 3x + 2y + 4 = 0 Or $y = \frac{-3x}{2} - 2$

Any line perpendicular to it will have slope = $\frac{2}{3}$

Thus equation of line through (2, 3) and slope $\frac{2}{3}$ is

(y – 3) = $\frac{2}{3}$(x – 2)

$\Rightarrow$3y – 9 = 2x – 4

$\Rightarrow$3y – 2x – 5 = 0.

Ex 26. Find the perpendicular distance between the parallel lines $3x + 4y + 3 = 0$ and $3x + 4y + 12 = 0$

Sol: The distance between parallel line $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$ is given by

$$= \frac{c_1 - c_2}{\sqrt{a^2 + b^2}} = \frac{12-3}{\sqrt{3^2 + 4^2}} = \frac{9}{5}$$

Ex 27. If (1,2), (4,5), (8,9) and (x, y) are the vertices of a paralelogram, then find the value of $x + y$.

Sol: Since the points (1, 2), (4, 5) and (8, 9) are collinear. Therefore, the given four points cannot form a parallelogram. Therefore the given data is inconsistent.

Ex 28. Find the area of the triangle formed by the vertices (4, 5), (10, 12) and (-3, 2)

Sol: The area of the triangle having its vertices as $P(x_1, y_1)$, $Q(x_2, y_2)$ and $R(x_3, y_3)$ is given by

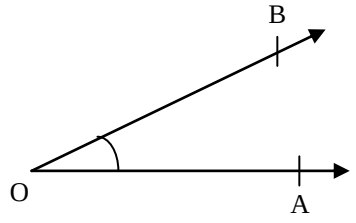
$$\frac{1}{2}\left[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right]$$

$$= \frac{1}{2}\left[4(12+2) + 10(-2-5) + (-3)(5-12)\right] = \frac{1}{2}\left[4.14 - 10.7 + (-3)(-7)\right] = \frac{1}{2}\left[56 - 70 + 21\right] = 3.5$$

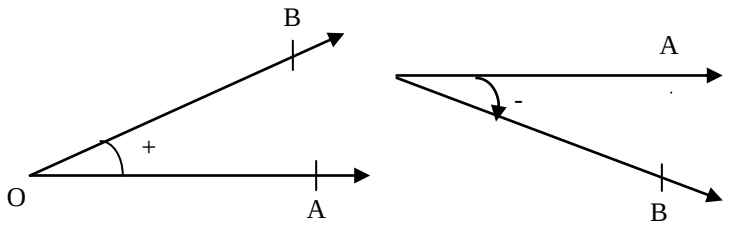
Trigonometry

3.1 Definitions & Review of Concepts

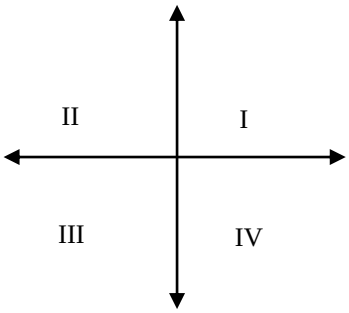
Angle: When a ray OA rotates about the point O from in initial position OA to final position OB, we say that ∠AOB has been formed.
OA is called the initial side and OB, the terminal side of ∠AOB.



Positive & Negative Angles: When the ray OA rotates in anticlockwise direction, a positive angle is formed, and when it moves in clockwise direction, a negative angle is formed.



Quadrants: Let *X' OX* and *YOY'* be two lines at right angles to each other. We call *X' OX* and *YOY'* as *x*-axis and *y*-axis respectively.
These lines divide the plane into 4 parts. *The parts XOY, YOX', X'OY'* and *Y'OX are known as* 1st, 2^{n}, 3rd *and* 4th *quadrant respectively.*
An angle is said to be in a particular quadrant, if the terminal side of the angle lies in that quadrant.

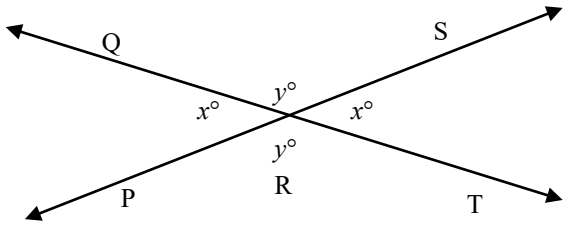


Measurement of Angles: Mainly there are three systems for measuring angles.
Sexagesimal system: In this system, angle is measured in degrees.
I *right angle* = 90^0, 1^0 = 60' and 1' = 60".
Centesimal system: In this system angle is measured in grades.
I *right angle* = 100^g, 1^g = 100' and 1' = 100".
Circular Measure: In this system, angle is measured in radius.
The angle subtended at the centre of a circle by an arc of length equal to its radius is 1 *radian, written as* 1^c.

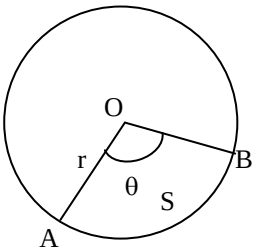
$\pi^c = 180^0 = 200^R$ = 2 right angles

3.2 Formule

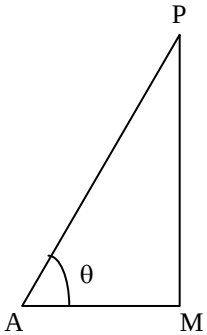
I. Relation between Angle, Radius & Arc Length:



If θ^C is the angle made by an arc of lengths at the centre of a circle of radius r, then $\theta = \frac{S}{r}$.



II. Trigonometric Ratios:



$$\sin\theta = \frac{\text{Perp.}}{\text{Hyp.}} = \frac{PM}{AP}; \cos\theta = \frac{\text{Base}}{\text{Hyp.}} = \frac{AM}{AP};$$

$$\tan\theta = \frac{\text{Perp.}}{\text{Base}} = \frac{PM}{AM};$$

$$\text{cosec}\,\theta = \frac{1}{\sin\theta}; \sec\theta = \frac{1}{\cos\theta}; \cot\theta = \frac{\cos\theta}{\sin\theta}.$$

III. Identities:

(i) $\tan\theta = \frac{\sin\theta}{\cos\theta}$ (ii) $\cot\theta = \frac{\cos\theta}{\sin\theta}$

(iii) $\sin^2\theta + \cos^2\theta = 1$ (iv) $1 + \tan^2\theta = \sec^2\theta$

(v) $1 + \cot^2\theta = \text{cosec}^2\theta$

IV. Values of T-Ratios:

Angle	$\sin\theta$	$\cos\theta$	$\tan\theta$
0^0	0	1	0
$30^0 = \frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
$45^0 = \frac{\pi}{4}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1
$60^0 = \frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$90^0 = \frac{\pi}{2}$	1	0	∞
$180^0 = \pi$	0	-1	0

V. Sign of T-Ratios:

1st Quadrant: All positive.
2nd Quadrant: sin θ & cosec θ positive.
3rd Quadrant: tan θ & cot θ positive.
4th Quadrant: cos θ & sec θ positive.

Remember:	**I**	**II**	**III**	**IV**
	All	**sin**	**tan**	**cos**

VI. Range of T-Ratios:

(i) $-1 \le \sin\theta \le 1$ and $-1 \le \cos\theta \le 1$
Thus, $|\sin\theta| \le 1$ and $|\cos\theta| \le 1$.
The value of sin θ *is never greater than 1 and never less than* – 1.
The value of cos θ *is never greater than 1 and never less than* – 1.
(ii) cosec $\theta \ge 1$ and cosec $\theta \le -1$.
(iii) sec $\theta \ge 1$ and sec $\theta \le -1$.
(iv) tan θ may assume any value.

VII. Increasing & Decreasing Functions:

(i) The value of sin θ increase from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$. As θ increases in this interval, then sin θ also increases.
(ii) 1st Quadrant: sin θ increases from 0 to 1 ; cos θ decreases from 1 to 0 and tan θ increases from 0 to ∞.
(iii) 2nd Quadrant: sin θ decreases from 1 to 0 ; cos θ decreases from 0 to – 1; tan θ decreases from ∞ to 0.
(iv) 3rd Quadrant: sin θ decreases from 0 to – 1 ; cos θ increases from – 1 to 0; tan θ increases from 0 to ∞.
(v) 4th Quadrant: sin θ increases from – 1 to 0; cos θ increases from 0 to 1; tan θ decreases from ∞ to 0.

VIII. T-Ratios of Negative, Complementary, Supplementary Angles etc.

(i) $\sin(-\theta) = -\sin\theta$, $\cos(-\theta) = \cos\theta$, $\tan(-\theta) = -\tan\theta$.
(ii) $\sin(90^0 - \theta) = \cos\theta$; $\cos(90^0 - \theta) = \sin\theta$; $\tan(90^0 - \theta) = \cot\theta$.
(iii) $\sin(90^0 + \theta) = \cos\theta$; $\cos(90^0 + \theta) = -\sin\theta$; $\tan(90^0 + \theta) = -\cot\theta$.
(iv) $\sin(180^0 - \theta) = \sin\theta$; $\cos(180^0 - \theta) = -\cos\theta$; $\tan(180^0 - \theta) = -\tan\theta$.
(v) $\sin(180^0 + \theta) = -\sin\theta$; $\cos(180^0 + \theta) = -\cos\theta$; $\tan(180^0 + \theta) = \tan\theta$.
(vi) $\sin(360^0 - \theta) = -\sin\theta$, $\cos(360^0 - \theta) = \cos\theta$; $\tan(360^0 - \theta) = -\tan\theta$.
(vii) $\sin(360^0 + \theta) = \sin\theta$, $\cos(360^0 + \theta) = \cos\theta$, $\tan(360^0 + \theta) = \tan\theta$.

IX. Sum & Difference Formula:

(i) $\sin(x + y) = \sin x \cos y + \cos x \sin y$.
(ii) $\sin(x - y) = \sin x \cos y - \cos x \sin y$.
(iii) $\cos(x + y) = \cos x \cos y - \sin x \sin y$
(iv) $\cos(x - y) = \cos x \cos y + \sin x \sin y$.
(v) $2 \sin x \cos y = \sin(x + y) + \sin(x - y)$.
(vi) $2 \cos x \sin y = (x + y) - \sin(x - y)$.
(vii) $2 \cos x \cos y = \cos(x + y) + \cos(x - y)$.
(viii) $2 \sin x \sin y = \cos(x - y) - \cos(x + y)$.
(ix) $\sin^2 x - \sin^2 y = \sin(x + y).\sin(x - y)$.
(x) $\cos^2 x - \sin^2 y = \cos(x + y).\cos(x - y)$.

X. Some More Formulae:

(i) sin C + sin D = 2 sin $\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$

(ii) sin C – sin D = 2 cos $\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$

(iii) cos C + cos D = 2 cos $\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$

(iv) cos C – cos D = - 2 sin $\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$

XI. Tangent Formulae:

(i) tan (x + y) = $\frac{\tan x+\tan y}{1-\tan x\tan y}$

(ii) tan (x – y) = $\frac{\tan x-\tan y}{1+\tan x\tan y}$

XII. T-Ratios of Multiple Angles:

(i) sin $2x$ = 2 sin x cos x
(ii) cos $2x$ = $\cos^2 x - \sin^2 x$ = 2 $\cos^2 x$ -1 = 1 – 2 $\sin^2 x$.
(iii) 1 + cos $2x$ = 2 $\cos^2 x$ and 1 – cos $2x$ = 2 $\sin^2 x$.
(iv) sin $3x$ = 3 sin x – 4 $\sin^3 x$
(v) cos $3x$ = 4 $\cos^3 x$ – 3 cos x

(vi) tan $2x$ = $\frac{2\tan x}{1-\tan^2 x}$

(vii) tan $3x$ = $\left(\frac{3\tan x-\tan^3 x}{1-3\tan^2 x}\right)$

(viii) cos $2x$ = $\frac{1-\tan^2 x}{1+\tan^2 x}$

(ix) sin $2x$ = $\frac{2\tan x}{1+\tan^2 x}$

XIII. T-Ratios of Sub-multiple Angles:

(i) sin x = 2 sin (x/2) cos (x/2)

(ii) cos x = $\cos^2\frac{x}{2}-\sin^2\frac{x}{2}=2\cos^2\frac{x}{2}-1=1-2\sin^2\frac{x}{2}$

(iii) tan x = $\frac{2\tan(x/2)}{1-\tan^2(x/2)}$

(iv) 1 – cos x = 2 $\sin^2\frac{x}{2}$ and 1 + cos x = 2 $\cos^2\frac{x}{2}$

(v) sin x = $\frac{2\tan(x/2)}{1+\tan^2(x/2)}$

(vi) cos x = $\frac{1-\tan^2(x/2)}{1+\tan^2(x/2)}$

XIV. T-Ratios of Some Special Angles:

(i) sin 15^0 = $\frac{\sqrt{3}-1}{2\sqrt{2}} = \cos 75^0$

(ii) cos 15^0 = $\frac{\sqrt{3}+1}{2\sqrt{2}} = \sin 75^0$

(iii) sin 18^0 = $\frac{\sqrt{5}-1}{4} = \cos 72^0$

(iv) cos 18^0 = $\frac{\sqrt{10+2\sqrt{5}}}{4} = \sin 72^0$

(v) sin 36^0 = $\frac{\sqrt{10-2\sqrt{5}}}{4} = \cos 54^0$

(vi) cos 36^0 = $\frac{\sqrt{5}+1}{4} = \sin 54^0$

(vii) sin 22 $\frac{1^0}{2}$ = $\frac{\sqrt{2-\sqrt{2}}}{2}$

(viii) cos 22 $\frac{1^0}{2}$ = $\frac{\sqrt{2+\sqrt{2}}}{2}$

XV. Trigonometric Equations:

(i) $\sin\theta = 0 \Rightarrow \theta = n\pi$

(ii) $\cos\theta = 0 \Rightarrow \theta = (2n+1)\frac{\pi}{2}$

(iii) $\tan\theta = 0 \Rightarrow \theta = n\pi$

XVI. Periodicity: *A function f (x) has periodicity p, if f (x + p) = f (x).* sin *x* has periodicity 2π, since sin (*x* + 2π) = sin *x*. cos *x* has periodicity 2π, since cos (*x* + 2π) = cos *x*

3.3 Things to Remember

1. $\frac{\pi}{6} = 30^0$, $\frac{\pi}{4} = 45^0$, $\frac{\pi}{3} = 60^0$, $\frac{\pi}{2} = 90^0$, $\pi = 180^0$ & $2\pi = 360^0$.

2. (i) All T-ratios are positive in 1st quadrant.
(ii) sin θ and cosec θ are + ve in 1st & 2nd qradrants.
(iii) tan θ and cot θ are + ve in 1st & 3rd quadrants.
(iv) cos θ and sec θ are + ve in 1st & 4th quadrants.

3. (i) sin ($180^0 - \theta$) = sin θ; tan ($180^0 + \theta$) = tan θ; cos ($360^0 - \theta$) = cos θ.
(ii) sin ($\pi - \theta$) = sin θ; tan ($\pi + \theta$) = tan θ, cos ($2\pi - \theta$) = cos θ.

4. (i) 2 sin *A* cos *B* = sin (*A* + *B*) + sin (*A* – *B*)
(ii) 2 cos *A* sin *B* = sin (A + B) – sin (*A* – *B*)
(iii) 2 cos *A* cos *B* = cos (*A* + *B*) + cos (*A* – *B*)
(iv) 2 sin *A* sin *B* = cos (*A* -*B*) – cos (*A* + *B*)

3.4 Solved Examples:

1. Convert $\frac{5\pi}{6}$ radians into degrees.

Sol. $\pi^c = 180^0$

$\therefore \left(\frac{5\pi}{6}\right)^c = \left(\frac{180}{\pi} \times \frac{5\pi}{6}\right)^0 = 150^0$

2. Convert 135^0 into radians.

Sol. $180^0 = \pi^c$

$\therefore 135^0 = \left(\frac{\pi}{180} \times 135\right)^c = \left(\frac{3\pi}{4}\right)$radians

Also, $\cos\theta = (\cot\theta \times \sin\theta) = \frac{9}{40} \times \left(\frac{-40}{41}\right) = \frac{-9}{41}$, $\sec\theta = \frac{1}{\cos\theta} = -\frac{41}{9}$.

3. Find the values of:

(i) $\sin 150^0$ (ii) $\cos 135^0$ (iii) $\tan(-210^0)$
(iv) $\cot 240^0$ (v) $\sec 330^0$ (vi) $\text{cosec } 390^0$

Sol. (i) $\sin 150^0 = \sin(180^0 - 30^0) = \sin 30^0 = \frac{1}{2}$

(ii) $\cos 135^0 = \cos(180^0 - 45^0) = -\cos 45^0 = -\frac{1}{\sqrt{2}}$

(iii) $\tan(-210^0) = -\tan 210^0 = -\tan(180^0 + 30^0) = -\tan 30^0 = -\frac{1}{\sqrt{3}}$

(iv) $\cot 240^0 = \cot(180^0 + 60^0) = \cot 60^0 = \frac{1}{\sqrt{3}}$

(v) $\sec 330^0 = \sec(360^0 - 30^0) = \sec 30^0 = \frac{1}{\cos 30^0} = \frac{2}{\sqrt{3}}$

(vi) $\text{cosec } 390^0 = \text{cosec}(360^0 + 30^0) = \text{cosec } 30^0 = \frac{1}{\sin 30^0} = 2$

4. If $\tan 62^0 = \frac{p}{q}$, find the value of $\tan 28^0$.

Sol. $\tan 28^0 = \tan(90^0 - 62^0) = \cot 62^0 = \frac{p}{q}$,

5. Evaluate:

(i) $\sin 79^0 \cos 19^0 - \cos 79^0 \sin 19^0$
(ii) $\cos 48^0 \cos 12^0 - \sin 48^0 \sin 12^0$
(iii) $\cos 70^0 \cos 40^0 + \sin 70^0 \sin 40^0$
(iv) $\sin 36^0 \cos 9^0 + \cos 36^0 \sin 9^0$

Sol. (i) $\sin 79^0 \cos 19^0 - \cos 79^0 \sin 19^0 = \sin (79^0 - 19^0) = \sin 60^0 = \frac{\sqrt{3}}{2}$ [$\because \sin x \cos y - \cos x \sin y = \sin (x - y)$]

(ii) $\cos 48^0 \cos 12^0 - \sin 48^0 \sin 12^0 = \cos (48^0 + 12^0) = \cos 60^0 = \frac{1}{2}$ [$\because \cos x \cos y - \sin x \sin y = \cos (x + y)$]

(iii) $\cos 70^0 \cos 40^0 + \sin 70^0 \sin 40^0 = \cos (70^0 - 40^0) = \cos 30^0 = \frac{\sqrt{3}}{2}$ [$\because \cos x \cos y + \sin x \sin y = \cos (x - y)$]

(iv) $\sin 36^0 \cos 9^0 + \cos 36^0 \sin 9^0 = \sin (36^0 + 9^0) = \sin 45^0 = \frac{1}{\sqrt{2}}$

6. Find the values of $\tan 15^0$ and $\cos 105^0$.

Sol. $\tan 15^0 = \tan (45^0 - 30^0) = \frac{\tan 45^0 - \tan 30^0}{1 + \tan 45^0 \tan 30^0} = \frac{1 - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}} = \left(\frac{\sqrt{3} - 1}{\sqrt{3} + 1}\right)$ $\cos 105^0 = \cos (60^0 + 45^0)$

$= \cos 60^0 \cos 45^0 - \sin 60^0 \sin 45^0 = \left(\frac{1}{2} \times \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}\right) = \left(\frac{1 - \sqrt{3}}{2\sqrt{2}}\right)$

7. Which of the following relations is true?

(i) $\sec^2 \theta + \tan^2 \theta = 1$ (ii) $\tan^2 \theta + \cot^2 \theta = 1$
(iii) $\text{cosec}^2 \theta - \cot^2 \theta = 1$ (iv) $\cot^2 \theta - \text{cosec}^2 \theta = 1$

Sol. Since $1 + \cot^2 \theta = \text{cosec}^2 \theta$, so $\text{cosec}^2 \theta - \cot^2 \theta = 1$ is the true statement.

8. If x and y are the angles lying in the second quadrant and $x < y$, then which one of the following is true?

(a) $\sin x < \sin y$ (b) $\sin x > \sin y$ (c) $\sin x = \sin y$

Sol. In 2nd quadrant $\sin x$ decreases as x increases.
$\therefore y > x \Rightarrow \sin y < \sin x$
Hence $x < y \Rightarrow \sin x > \sin y$

9. If $\sin \theta = \frac{4}{5}$, find the values of $\sin 2\theta$, $\cos 2\theta$, $\tan 2\theta$, $\sin 3\theta$ and $\cos 3\theta$

Sol. $\cos \theta = \sqrt{1 - \sin^2 \theta} = \sqrt{1 - \frac{16}{25}} = \sqrt{\frac{9}{25}} = \frac{3}{5}$

$\therefore \sin 2\theta = 2 \sin \theta \cos \theta = \left(2 \times \frac{4}{5} \times \frac{3}{5}\right) = \frac{24}{25}$;

$\cos 2\theta = (1 - 2 \sin^2 \theta) = \left(1 - 2 \times \frac{16}{25}\right) = \frac{-7}{25}$;

$\tan 2\theta = \frac{\sin 2\theta}{\cos 2\theta} = \frac{24}{25} \times \left(\frac{25}{-7}\right) = \frac{-24}{7}$;

$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta = \left[3 \times \frac{4}{5} - 4 \times \left(\frac{4}{5}\right)^3\right] = \frac{44}{125}$;

$\cos 3\theta = (4\cos^3\theta - 3\cos\theta) = \left[4\times\left(\frac{3}{5}\right)^3 - 3\times\frac{3}{5}\right] = -\frac{117}{125}$;

10. If $\tan\phi = \frac{3}{4}$, find the values of $\cos 2\phi$ and $\sin 2\phi$.

Sol. $\cos 2\phi = \frac{1-\tan^2\phi}{1+\tan^2\phi} = \left(\frac{1-\frac{9}{16}}{1+\frac{9}{16}}\right) = \left(\frac{7}{16}\times\frac{16}{25}\right) = \frac{7}{25}$

$\sin 2\phi = \frac{2\tan\phi}{1+\tan^2\phi} = \left(\frac{2\times\frac{3}{4}}{1+\frac{9}{16}}\right) = \left(\frac{3}{2}\times\frac{16}{25}\right) = \frac{24}{25}$

11. Evaluate: $\cos^2 45^0 - \sin^2 15^0$.

Sol. We know that $\cos^2\theta - \sin^2\phi = \cos(\theta+\phi) \,.\, \cos(\theta-\phi)$

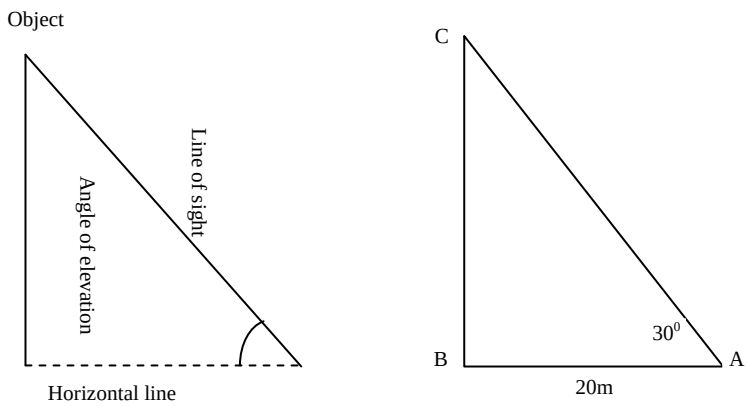
$\therefore \cos^2 45^0 - \sin^2 15^0 = \cos(45^0 + 15^0) \,.\, \cos(45^0 - 15^0)$

$= \cos 60^0 \,.\, \cos 30^0 = \phi = \frac{1}{2}\times\frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}$

3.5 Angles of Elevation and Depression

Suppose we wish to determine the height of a tall tree without climbing to the top of it. We could stand on the ground at a point some distance (say 20m) from the foot B of the tree.

Suppose we are able to measure angle BAC and suppose we find it to be 30^0. Then, just as in Ex, we calculate the height BC of the tree to be BC= $\frac{20}{\sqrt{3}}$ =11.5m (approx)



Suppose we are viewing an object. The line of sight or the line of vision is a straight line from our eye to the object we are viewing.
If the object is above the horizontal form the eye (i.e. if it is at object. In the process, our eyes move through an angle. This angle is called the angle of elevation of the object.
If the object is below the horizontal from the eye (i.e., at a lower level than ourselves), then we have to turn our head downwards to view the object. In the process, our eyes move through an angle. This angle is called the angle of depression of the object.

12. A man wishes to find the height of a fagpost which stand on a elevation of the top of the flagpost to be 45^0. On walking 30 metres towards the tower he find the corresponding angle of elevation to be 60^0. Find the height of the flagpost.

1. 62m 2. 82m 3. 71m 4. $30\sqrt{3}$m 5. None of these

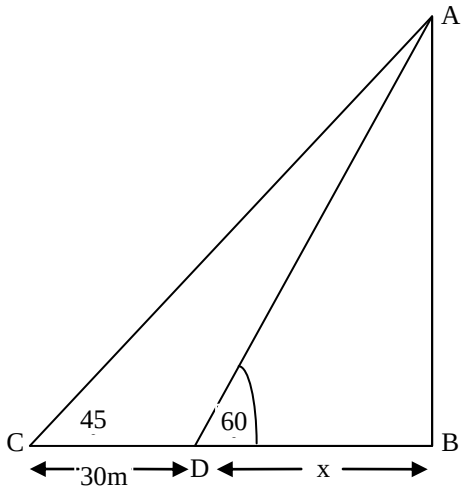
Sol. AB=height of flagpost = xm

In Δ ABD

Tan $60^0 = \frac{AB}{BD}$

$\therefore$ BD= $\frac{x}{\sqrt{3}}$.....(i)

tan 45^0 = $\frac{AB}{BD+DC}$



$\therefore \frac{x}{\sqrt{3}} + 30 = x \quad \therefore \frac{x(\sqrt{3}-1)}{\sqrt{3}} = 30 \quad \therefore x = \frac{30\sqrt{3}}{0.732} \approx 71m$

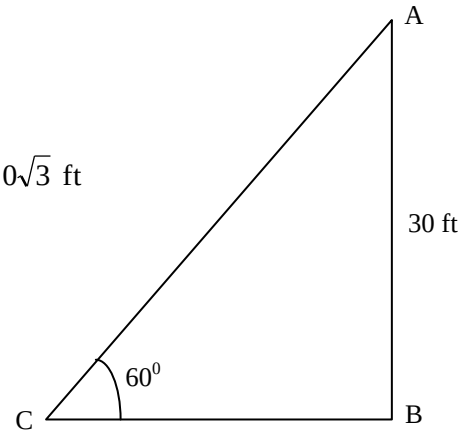
13. A small boy is standing at some distance from a flagpost. When he sees the flag the angle of elevation formed is 60^0. If the height of the flagpost is 30ft, what is the distance of the child from the flagpost?

1. $15\sqrt{3}$ ft 2. $10\sqrt{3}$ ft 3. $20\sqrt{3}$ ft 4. $\frac{20}{\sqrt{3}}$ ft 5. None of these

Sol. $\frac{AB}{BC} = \tan 60^0$

Or, $\frac{30}{BC} = \sqrt{3}$

Or, BC= $\frac{\sqrt{3}\times\sqrt{3}\times 10}{\sqrt{3}} = 10\sqrt{3}$ ft



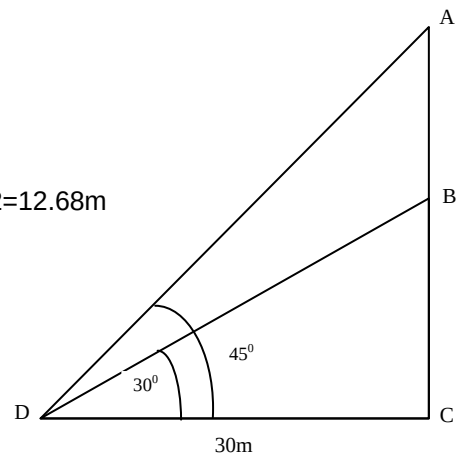
14. The angles of elevation of top and bottom of a flag kept on a flagpost from 30 metres distance are 45^0 and 30^0 respectively. What is the height of the flag?

1. 17.32m 2. 14.32m 3. 12.68m 4. $12\sqrt{3}$ ft 5. None of these

Sol. tan 45^0 = $\frac{AC}{30}$ or, $AC = 30m$

tan 30^0 = $\frac{BC}{30}$ or, $BC = \frac{30}{\sqrt{3}}$

Height of flag AB= 30 $-\frac{30}{\sqrt{3}} = 30 - 10\sqrt{3}$ = 30-17.2=12.68m

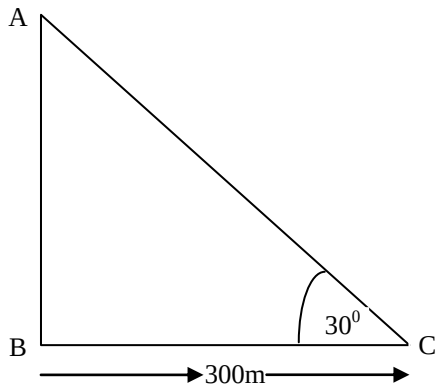


15. 300m from the foot of a cliff on level ground, the angle of elevation of the top of a cliff is 30^0. Find the height of this cliff.

Sol. Let the height of the cliff AB be x m
In Δ ABC

tan 30^0 = $\frac{AB}{BC} = \frac{x}{300}$

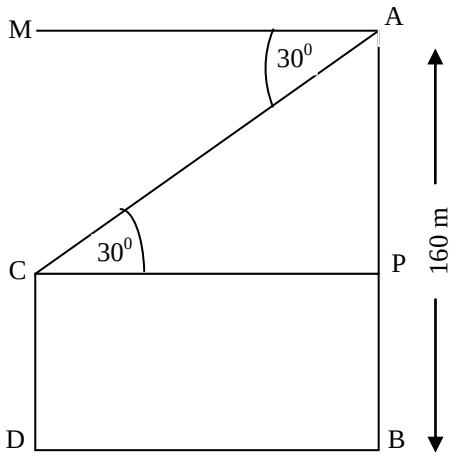
$\therefore = \frac{300}{\sqrt{3}} = 100\sqrt{3} - 173.20m$



16. The horizontal distance between two towers is $50\sqrt{3}$ m. The angle of depression of the first tower when seen from the top of the second tower is 30^0. If the height of the second tower is 160m, find the height of the first tower.

Sol. Let AB be the tower 160m high.
Let CD be another tower of height x m
Since, AM 33 PC
$\therefore$ angle MAC = angle ACP = 30^0
So, in Δ APC

tan 30^0 = $\frac{AP}{PC} \Rightarrow \frac{1}{\sqrt{3}} = \frac{AP}{50\sqrt{3}}$



$\therefore$ AP = 50m
$\therefore$ The height of the other tower – AB-AP=160-50=110m